

GIANTS CASTLE, South Africa

WMO: 685890

Lat: 29.2650S

Lon: 29.5222E

Elev: 1762

StdP: 81.87

Time Zone: 2.00 (E02)

Period: 98-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	0.5	1.9	-18.2	0.9	11.9	-14.8	1.3	12.0	10.2	11.8	8.2	12.5	1.8	180	0.419

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	8.5	28.1	14.9	26.8	14.8	25.5	14.8	18.7	23.9	18.1	23.0	17.4	22.1	4.0	20

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
17.1	15.2	20.9	16.5	14.6	20.2	15.9	14.0	19.6	61.5	24.1	59.0	23.1	56.7	22.2	21.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 7.5	6.3	5.4	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			DB													
			WB	-4.4	20.1	1.3	0.7	-5.4	20.6	-6.1	21.1	-6.8	21.5	-7.7	22.0	(4)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	14.8	17.9	17.5	17.0	14.9	13.0	10.9	10.5	12.7	14.4	15.1	15.8	17.7	(6)	
	DBStd	4.12	2.54	2.77	2.81	3.09	3.05	2.75	3.14	3.58	4.64	4.08	3.64	2.94	(7)	
	HDD10.0	107	0	1	1	3	9	22	32	15	15	7	3	0	(8)	
	HDD18.3	1441	39	43	58	107	166	223	243	177	130	116	91	47	(9)	
	CDD10.0	1845	244	209	217	151	103	48	48	99	147	164	176	238	(10)	
	CDD18.3	137	25	19	17	5	1	0	0	2	12	14	14	27	(11)	
	CDH23.3	927	143	82	74	28	6	1	1	17	124	135	126	189	(12)	
	CDH26.7	129	21	7	5	2	0	0	0	1	16	21	21	38	(13)	
(14) Wind	WSAvg	2.3	1.9	1.9	2.0	2.0	2.4	2.7	2.8	2.7	2.7	2.3	2.2	2.1	(14)	
(15) Precipitation	PrecAvg	1043	200	160	145	70	17	13	6	31	50	84	124	140	(15)	
	PrecMax	1375	396	270	327	153	54	100	22	145	135	182	265	259	(16)	
	PrecMin	766	40	74	35	31	0	0	0	0	0	30	77	64	(17)	
	PrecStd	169	84	60	77	33	17	25	7	38	43	39	48	55	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	29.0	27.8	27.2	26.3	24.2	21.9	21.7	25.4	28.7	28.9	29.0	29.9	(19)	
		MCWB	16.6	16.5	16.2	15.4	11.4	8.3	7.4	10.5	11.7	13.4	14.8	16.2	(20)	
	2%	DB	26.6	25.8	25.4	23.9	21.9	19.8	19.6	22.9	26.5	26.8	26.4	27.2	(21)	
		MCWB	16.9	17.1	16.3	14.0	10.5	7.7	7.1	9.5	11.1	12.9	14.7	16.4	(22)	
	5%	DB	24.8	24.1	23.5	21.9	20.2	18.0	17.9	21.0	24.7	24.7	24.5	25.2	(23)	
		MCWB	16.9	17.1	16.2	13.7	10.0	7.2	6.7	8.6	10.7	12.7	14.8	16.2	(24)	
	10%	DB	22.9	22.3	21.8	20.1	18.4	16.2	16.1	19.1	22.7	22.5	22.5	23.3	(25)	
		MCWB	16.4	16.8	15.8	13.2	9.5	6.7	6.1	8.2	10.2	12.4	14.6	16.1	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	19.4	19.4	19.4	17.4	13.6	10.3	9.6	12.2	14.4	16.6	17.7	19.6	(27)	
		MCDB	25.2	24.3	23.7	22.3	19.9	17.8	17.7	21.7	23.1	22.2	24.2	25.9	(28)	
	2%	WB	18.4	18.5	18.1	16.2	12.1	9.1	8.4	10.8	13.1	15.3	16.7	18.4	(29)	
		MCDB	23.6	23.3	22.6	21.1	18.9	16.4	15.7	19.4	21.5	21.7	22.7	23.7	(30)	
	5%	WB	17.7	17.8	17.2	15.1	11.1	8.3	7.5	9.9	12.1	14.5	16.0	17.5	(31)	
		MCDB	22.4	22.4	21.5	19.6	17.8	15.6	14.9	17.6	20.3	20.6	21.8	22.5	(32)	
	10%	WB	17.1	17.1	16.2	14.0	10.3	7.4	6.8	9.1	11.4	13.7	15.2	16.8	(33)	
		MCDB	21.4	21.0	20.3	18.8	16.9	14.3	14.2	16.3	19.1	19.2	20.7	21.4	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	8.5	8.3	8.8	9.8	11.3	12.1	11.9	12.0	11.9	10.1	9.7	9.2	(35)	
		MCDBR	12.3	11.6	11.9	12.3	13.0	13.8	13.3	13.8	14.9	14.8	14.1	13.4	(36)	
	5% WB	MCWBR	6.1	5.5	5.6	5.5	6.0	7.0	6.5	5.8	5.5	5.6	5.9	6.0	(37)	
		MCWBR	10.2	9.7	9.7	10.2	11.6	12.4	12.1	12.5	12.9	11.8	12.1	10.9	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.370	0.368	0.352	0.310	0.280	0.261	0.268	0.329	0.376	0.403	0.363	0.366	(40)	
		taud	2.365	2.376	2.395	2.483	2.539	2.560	2.523	2.351	2.236	2.204	2.350	2.366	(41)	
	Ebn at Noon		971	955	938	931	919	918	923	894	897	910	971	977	(42)	
		Edh at Noon	132	126	117	96	81	75	81	108	135	148	133	132	(43)	
(44) All-Sky Solar Radiation	RadAvg		6.14	5.84	5.21	4.42	3.85	3.49	3.77	4.39	5.20	5.54	6.12	6.24	(44)	
	RadStd		0.42	0.49	0.33	0.27	0.20	0.20	0.20	0.26	0.31	0.43	0.41	0.37	(45)	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (10 neighbors)	+0.33	+0.57	N/A	N/A	-0.34	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page